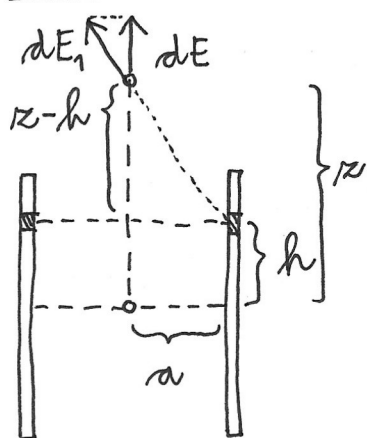


EMP, 1. A PISNI IZPIT

1 NABITA IZOLATORSKA CEV



rešujemo za primer $r \geq 0$

a) $\left\{ \begin{aligned} dE_1 &= \frac{dq}{4\pi\epsilon_0 [(r-h)^2 + a^2]} \end{aligned} \right. , dq = 5 dh a d\varphi$

$\left\{ \begin{aligned} \frac{dE}{dE_1} &= \frac{r-h}{\sqrt{(r-h)^2 + a^2}} \end{aligned} \right.$

$dE = \frac{5a}{4\pi\epsilon_0} \frac{r-h}{[(r-h)^2 + a^2]^{\frac{3}{2}}} dh d\varphi$

$\left\{ \begin{aligned} E &= -\frac{5a}{4\pi\epsilon_0} \int_0^{2\pi} d\varphi \int_{r+a}^{r-a} \frac{r-h}{[(r-h)^2 + a^2]^{\frac{3}{2}}} d(r-h) \end{aligned} \right. , (r-h)d(r-h) = \frac{1}{2} d(r-h)^2$

$\left\{ \begin{aligned} E &= -\frac{5a}{4\pi\epsilon_0} 2\pi \frac{1}{2} (-2) \frac{1}{\sqrt{(r-h)^2 + a^2}} \Big|_{r-h=r+a}^{r-a} \\ E &= \frac{5a}{2\epsilon_0} \left[\frac{1}{\sqrt{a^2 + (r-a)^2}} - \frac{1}{\sqrt{a^2 + (r+a)^2}} \right] \end{aligned} \right.$

b) $r \ll a$: $a^2 + (r \pm a)^2 = 2a^2 \pm 2ar + r^2 = 2a^2 \left(1 \pm \frac{r}{a} + \frac{r^2}{2a^2} \right)$

$\left\{ \begin{aligned} E &= \frac{5a}{2\epsilon_0} \frac{1}{\sqrt{2a^2}} \left[\left(1 - \frac{r}{a} \right)^{-\frac{1}{2}} - \left(1 + \frac{r}{a} \right)^{-\frac{1}{2}} \right] \approx \frac{5}{2\sqrt{2}\epsilon_0} \left[1 + \frac{1}{2} \frac{r}{a} - \left(1 - \frac{1}{2} \frac{r}{a} \right) \right] \end{aligned} \right.$

$\left\{ \begin{aligned} E &= \frac{5}{2\sqrt{2}\epsilon_0 a} r \end{aligned} \right.$

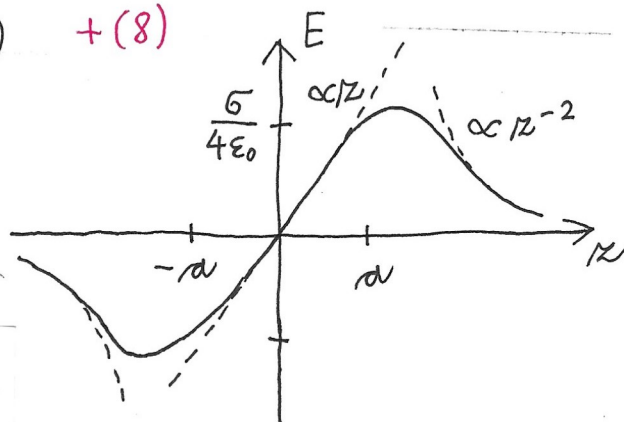
$\left\{ \begin{aligned} r \gg a : a^2 + (r \pm a)^2 &= \\ &= r^2 \left(1 \pm 2 \frac{a}{r} + 2 \frac{a^2}{r^2} \right) \end{aligned} \right.$

$\left\{ \begin{aligned} E &= \frac{5a}{2\epsilon_0} \frac{1}{\sqrt{r^2}} \left[\left(1 - 2 \frac{a}{r} \right)^{-\frac{1}{2}} - \left(1 + 2 \frac{a}{r} \right)^{-\frac{1}{2}} \right] \end{aligned} \right.$

$\left\{ \begin{aligned} E &= \frac{5a^2}{\epsilon_0} \cdot \frac{1}{r^2} \end{aligned} \right.$

8+ → 1

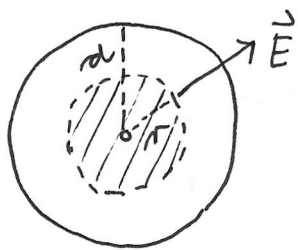
c) + (8)



$E(r=a) \approx 0.28 \frac{5}{\epsilon_0}$

$E(r) \rightarrow \text{LIHA}$

2. ELEKTRIČNA SILA V NABITEM VALJU



$$\rho = \frac{\alpha}{r}$$

$$+ (1) \left\{ \begin{aligned} q(r) &= \int_0^r \rho(r') 2\pi r' l dr' = \\ &= \int_0^r 2\pi \alpha l dr' = \underline{2\pi \alpha l \cdot r} \end{aligned} \right.$$

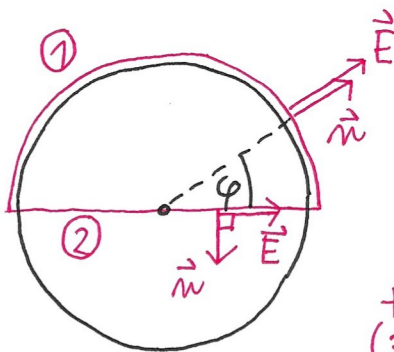
naboj v valju s polmerom $r < a$

+ (2) Gaussov izrek:
(za valj polmera r)

$$q(r) = \epsilon_0 E(r) \cdot S(r)$$

$$2\pi \alpha l \cdot r = \epsilon_0 E \cdot 2\pi r l \Rightarrow \underline{E(r) = \frac{\alpha}{\epsilon_0}}$$

konstantno polje v radialni smeri



$$\vec{F}_e = \epsilon_0 \oint \left[\vec{E}(\vec{E} \cdot \vec{n}) - \frac{1}{2} E^2 \vec{n} \right] dS$$

$$+ (3) \left\{ \begin{aligned} \textcircled{1} \quad \underline{\vec{E} \parallel \vec{n}} &\Rightarrow \vec{E}(\vec{E} \cdot \vec{n}) = \vec{E}E = E^2 \vec{n} \\ \vec{F}_{e1} &= \frac{\epsilon_0}{2} \int E^2 \vec{n} dS \end{aligned} \right.$$

$$+ (4) \left\{ \vec{n} = \begin{bmatrix} \cos \varphi \\ \sin \varphi \end{bmatrix}, dS = l a d\varphi \Rightarrow \vec{F}_{e1} = \frac{\epsilon_0}{2} \frac{\alpha^2}{\epsilon_0^2} l a \int_0^\pi \begin{bmatrix} \cos \varphi \\ \sin \varphi \end{bmatrix} d\varphi$$

$$+ (5) \left\{ \vec{F}_{e1} = \frac{\alpha^2}{2\epsilon_0} l a \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \underline{\frac{\alpha^2}{\epsilon_0} l a \hat{e}_y}$$

$$+ (6) \left\{ \textcircled{2} \quad \underline{\vec{E} \perp \vec{n}} \Rightarrow \vec{F}_{e2} = -\frac{\epsilon_0}{2} \int E^2 \vec{n} dS = -\frac{\epsilon_0}{2} E^2 \vec{n} \int dS$$

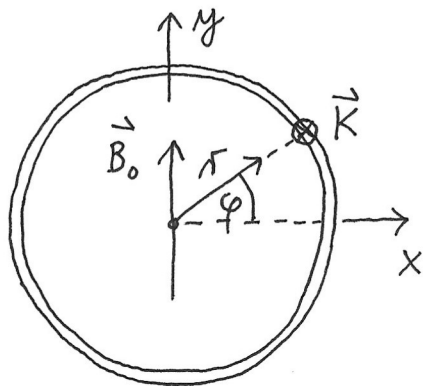
$$+ (7) \left\{ \vec{F}_{e2} = -\frac{\epsilon_0}{2} \frac{\alpha^2}{\epsilon_0^2} (-\hat{e}_y) 2al = \underline{\frac{\alpha^2}{\epsilon_0} l a \hat{e}_y}$$

$$+ (8) \left\{ \vec{F}_e = \vec{F}_{e1} + \vec{F}_{e2} = \frac{2\alpha^2}{\epsilon_0} al \hat{e}_y \Rightarrow \boxed{\frac{F_e}{l} = \frac{2\alpha^2 a}{\epsilon_0}}$$

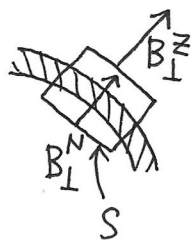
8+ → 1

karže NAVZGOR

3. MAGNETNO POLJE PREVODNE CEVI Z ELEKTRIČNIM TOKOM,

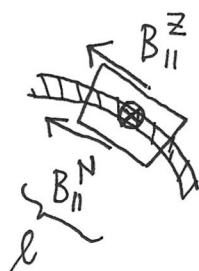


RP1 majhen kvader okoli stene cevi (prispevata le ZGORNJA in SPODNJA ploskev)



$$\begin{aligned} \vec{\nabla} \cdot \vec{B} &= 0 \\ \oint \vec{B} \cdot d\vec{S} &= \oint B_{\perp} dS = 0 \\ B_{\perp}^z S - B_{\perp}^N S &= 0 \\ \boxed{B_{\perp}^z = B_{\perp}^N} \end{aligned} \quad (1)$$

RP2 majhna pravokotna ranka okoli stene cevi (prispevata le ZGORNJA in SPODNJA stranica)



$$(2) \quad \vec{\nabla} \times \vec{B} = \mu_0 \vec{j} \Rightarrow \oint \vec{B} \cdot d\vec{l} = \oint B_{\parallel} dl = \mu_0 I$$

$$(3) \quad -B_{\parallel}^z l + B_{\parallel}^N l = \mu_0 I \Rightarrow \boxed{B_{\parallel}^N - B_{\parallel}^z = \mu_0 K}, \quad K = \frac{I}{l}$$

$$\vec{B} = -\vec{\nabla} U_M, \quad U_M^N = -B_0 y = -B_0 r \sin \varphi \quad \text{podano} \quad (4)$$

$$(5) \quad \left\{ \begin{aligned} U_M^z &= \frac{C}{r} \sin \varphi \end{aligned} \right. \quad \leftarrow \text{biti mora isti sin in } \varphi \text{ kot enostraj, da se robni pogoji izidejo, pa } r^m \rightarrow r^{-m}$$

U_M NI ZVEZEN, pač pa je $B_{\perp}$

$$\begin{aligned} (6) \quad \left\{ \begin{aligned} \text{RP1} \quad B_{\perp} &= -\frac{\partial}{\partial r} U_M \\ -\frac{C}{a^2} \sin \varphi &= -B_0 \sin \varphi \Rightarrow \boxed{C = B_0 a^2} \Rightarrow U_M^z = \frac{B_0 a^2}{r} \sin \varphi \end{aligned} \right. \end{aligned}$$

NASPROTNI predznak kot U_M^N !

$$\text{RP2} \quad \left\{ \begin{aligned} K &= \frac{1}{\mu_0} (B_{\parallel}^N - B_{\parallel}^z), \quad B_{\parallel} = -\frac{1}{r} \frac{\partial}{\partial \varphi} U_M \end{aligned} \right. (7)$$

$$(8) \quad \left\{ \begin{aligned} K &= \frac{1}{\mu_0} (B_0 \cos \varphi + \frac{B_0 a^2}{a^2} \cos \varphi) \\ \boxed{K(\varphi) = \frac{2B_0}{\mu_0} \cos \varphi} &\rightarrow \text{največja gostota toka je na levem in desnem delu stoda} \end{aligned} \right. \quad 8+ \rightarrow \boxed{1}$$